AMAYA KEYS

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EDUCATION

Howard University *Washington, D.C*
B.S., Computer Science; GPA: 3.97 *Expected May 2027*
Karsh STEM Scholars Program, Cohort 7

RESEARCH EXPERIENCE

Howard University Institute for Human-Centered AI *Washington, D.C.*
AI for Positive Change Lab, *Undergraduate Researcher* *Sept. 2025 - Present*
Advisors: Saurav Aryal, Gloria Washington

- Developing a multilingual NLP system for the SemEval 2027 StereoQueerEval shared task to classify LGBTQIA+-related stereotypes and hate speech in YouTube comments across English, Italian, Dutch, and Persian.
- Fine-tuned a 3.4B-parameter DeepSeek-OCR 2 vision-language model using supervised fine-tuning (SFT) and LoRA for scientific figure classification, data extraction, summarization, and visual question answering.
- Apply Python programming, prompt engineering, and machine learning techniques to evaluate the feasibility of employing multimodal LLMs for an AI-enabled child monitoring system.
- Assessed Qwen2-VL and VideoLLaVa for their accuracy in detecting child safety risks within video input compared to human perception.
- Labeled over 240 LLM prompt and response pairs and 400 generated images to evaluate clarity, groundedness, usability, and integrity.

NSF Applied AI for Advanced Applications REU Program *Worcester, MA*
Worcester Polytechnic Institute, *Summer Research Intern* *May – July 2026*
Advisor: Fabricio Murai

- Experimented with various machine learning techniques to predict food spoilage and shelf-life degradation using time-series data.
- Employed a mechanistic-neural hybrid modeling approach for projecting microbial growth curves using environmental conditions.
- Developed a chatbot interface for non-food safety experts to run and receive interpretable explanations of predictions.

NSF Pediatric Prosthetics REU Program *Orlando, FL*
University of Central Florida, *Summer Research Intern* *May – July 2025*
Advisors: Albert Manero, Peter Smith

- Developed an occupational therapy-adapted virtual reality game in Unity3D (C#) for children with a limb difference.
- Designed an experimental study to assess the game's usability and associated levels of engagement and motivation.
- Completed Responsible Conduct of Research (RCR) coursework on Biomedical and Human Subjects Research and Communicating Research Findings.

NSF Immersive 3D Visualization REU Program *New York, NY*
CUNY Hunter College, *Summer Research Intern* *June – July 2024*
Advisors: Oyewole Oyekoya, Daniel Chan

- Developed a virtual reality multi-sensory environment for adolescents with autism using Unity3D (C#).

- Designed an experimental study to measure how the environment's interactive features contribute to anxiety reduction and user satisfaction.
- Completed Responsible Conduct of Research (RCR) coursework on Social and Behavioral research.

- Presented research findings at the ACM Interactive Surfaces and Spaces '24 poster session.

PUBLICATIONS & PRESENTATIONS

A. E. Keys*, S. Phillips*, S. Gokulakrishnan*, P. Antala, T. B. Arnold, F. Murai. "Predicting Food Spoilage with Machine Learning," poster accepted for presentation at the Annual Biomedical Research Conference for Minoritized Scientists, Seattle, WA. November 19, 2026.

A. E. Keys*, S. Phillips*, S. Gokulakrishnan*, P. Antala, T. B. Arnold, F. Murai. "Neuromechanistic Shelf-Life Prediction with an Agentic, Explanation-Oriented Interface," paper presented at the MIT Undergraduate Research Technology Conference, Cambridge, MA. October 10, 2026. ***[To be published in the IEEE Xplore conference proceedings]***

A. E. Keys, S. K. Aryal, G. Washington. "Assessing Video LLM Performance in Detecting Child Safety Risks," poster presented at the Association of Computer Science Departments at Minority Institutions Symposium, Orangeburg, SC, USA. March 27-28, 2026. [***Awarded First Place Poster***]

A. E. Keys, A. D. V. Cruz, V. Rivera, P. Smith, M. Dombrowski, J. Sparkman, D. Turgut, A. Manero. "Gamifying Occupational Therapy for Prosthetic Training," poster presented at the University of Central Florida Summer Research Poster Showcase, Orlando, FL, USA. July 18, 2025.

A. E. Keys, O. Oyekoya, and D. Chan. "Exploring Effects of Interactive Virtual Reality Sensory Environment on Anxiety Reduction in Adolescents with Autism," Companion Proceedings of the 2024 Conference on Interactive Surfaces and Spaces, pp. 48–53. Poster presented at ACM Interactive Surfaces and Spaces, Vancouver, BC, CA. October 27, 2024. https://doi.org/10.1145/3696762.3698051

A. E. Keys*, A. Pierre*, Z. Francis*, S. Wright*. "DNA Barcoding for Exploration of Taxonomic Diversity and Ecological Management of Ant Species in Washington, D.C.," poster presented at the Howard University Research Symposium, Washington, DC, USA. April 27, 2023.

LEADERSHIP & SERVICE

Black Eudaimonia
President *June 2026 - Present*

- Direct and facilitate overall organization operations in collaboration with executive team to uphold our mission of centering Black joy in the pursuit of outdoor equity.

Social and Wellness Chair *Aug 2025 - May 2026*

- Led wellness events and initiatives to promote the collective health and well-being of all members.

Membership Chair *Oct 2024 – Aug 2025*

- Managed recruitment, retention and engagement efforts for new and existing members.

Prime Incubator Program
Fellow *Jan – Mar 2026*

- Engaged in an 8-week series of entrepreneurship workshops for HBCU innovators, developing key skills in marketing, sales, financial literacy, and business strategy.
- Participated in the 2026 Howard University Empower Pitch Competition.

Brown Kids Read
University Ambassador *Nov 2025 – May 2026*

- Conducted weekly storytime readings at a local elementary school, supporting literacy access and distributing free, diverse books to K-1st grade students.
- Collaborated with fellow ambassadors across the country to promote equitable storytelling and community-centered content.

Boolean Girl
Lead Instructor *Oct 2024 – Dec 2025*

- Created and delivered engaging coding lessons to elementary school-aged girls using Scratch and Turtle graphics.
- Led Saturday clubhouses that foster creativity, problem-solving skills and confidence in STEM.

Jumpstart College Corps Program
Corps Member & Team Lead *Sept 2024 – May 2025*

- Accrued over 300 service hours supporting preschool-aged children in strengthening kindergarten literacy skills through interactive books, vocabulary lessons, and theme-based activities.
- Completed Americorps trainings in language development and conflict resolution to enhance teaching effectiveness.

Howard University Alternative Spring Break (HU-ASB)
Participant, Chicago, IL *Mar 2025*

- Tutored 7th and 8th grade students in mathematics.
- Created snack and hygiene kits for youth experiencing homelessness.
- Built dirt beds for a community farm preparing for their upcoming harvest season.

Participant, Louisville, KY *Mar 2024*

- Organized community donations at The Salvation Army, Angel Tree warehouse, and local community garden.
- Assisted elementary school teachers with after-school administrative tasks.

National Capital Area Blacks at Microsoft (BAM) Academy
Assistant Instructor *Oct 2023 – Apr 2024*

- Assisted Microsoft employees with teaching grade school children beginner computing and programming principles.

MEMBERSHIP

Tau Beta Pi Engineering Society, DC Alpha Chapter *2025 - Present*
Society of Women Engineers *2023 - Present*

CERTIFICATIONS

PCAP-31-03: Certified Associate in Python Programming *May 2023*
OpenEDG Python Institute

98-381: Introduction to Programming Using Python *May 2022*
Microsoft Technical Associate (MTA)